

NLP Lab: Autogen

Progress Update 09.05.24

What was done

Conducted experiments:

- RAG: question answering
- ALFWorld: decision making in a text world environment
- Group chats: test dynamic group chats on 12 handcrafted tasks
- Multi-agent Coding: 3 out of 5 datasets

Streamlit demos:

- RAG (with and w/o interactive retrieval)
- Chess (with and w/o the dedicated board agent)
- Group chats

Deployment in Streamlit community cloud does not work (allegedly due to Docker), we will deploy on our own VM

Problems

- The library gets updated a lot: should we stick closer to the code used in the paper or to the up-to-date methods?
- Our results sometimes differ a lot from the ones in the paper (some additional attention might be needed here)
- Different dataset loading problems in the experiments

Next steps

- perform Online Decision Making for Browser interactions experiment
- integrate function calls in streamlit demo

StateFlow: Enhancing LLM Task-Solving through StateDriven Workflows
(<https://arxiv.org/pdf/2403.11322>):

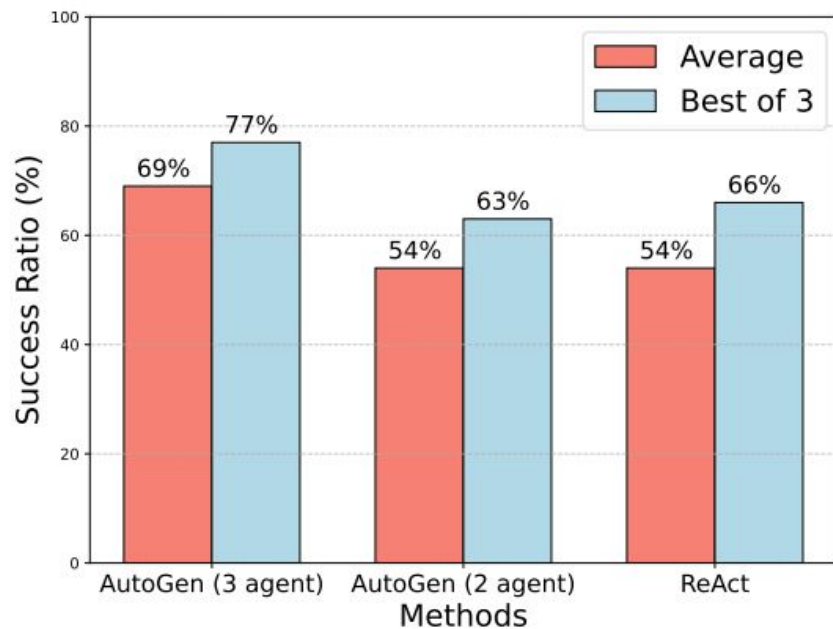
“Notably, we show that StateFlow can be combined with Reflexion (Shinn et al., 2023), improving the success rate on ALFWorld from 84.3% to 94.8% after 6 iterations.”

=> Use the state driven Workflow in the ALFWorld experiment

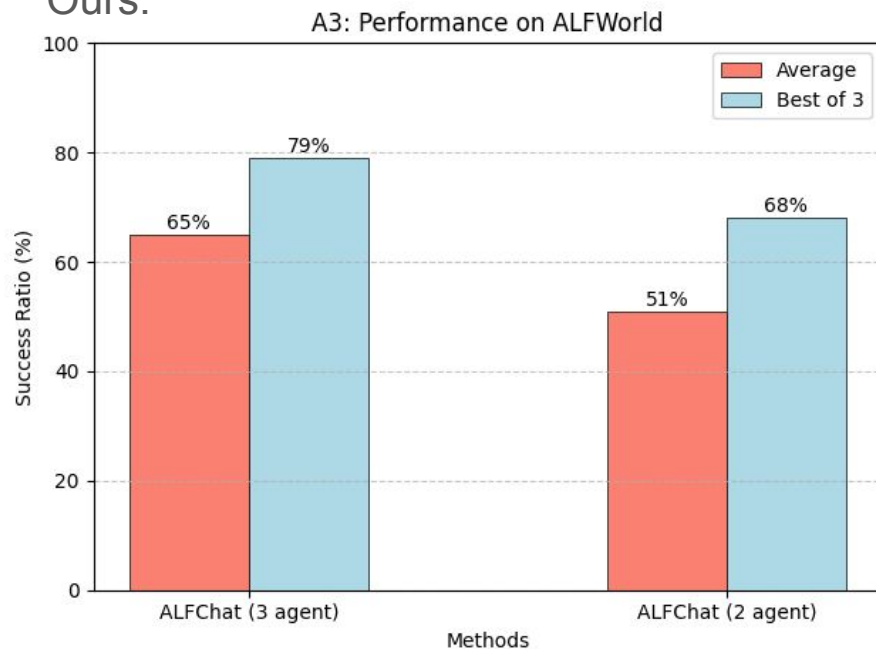
Try to come up with additional experiments not presented in the paper

Appendix: ALFWorld

Original:



Ours:



Appendix: Group chat

Original:

Table 5: Number of successes on the 12 tasks (higher the better).

Model	Two Agent	Group Chat	Group Chat with a task-based speaker selection policy
GPT-3.5-turbo	8	9	7
GPT-4	9	11	8

Ours:

Model	Two Agent	Group Chat
GPT-3.5-Turbo	6	8
GPT-4	9	7

Appendix: Multi-agent Coding

Original:

Dataset	netflow	facility	tsp	coffee	diet
Saving Ratio	3.14x (0.65)	3.14x (0.64)	4.88x (1.71)	3.38x (0.86)	3.03x (0.31)

Ours:

Dataset	tsp	coffee	diet
Saving Ratio	4.65x (1.58)	3.5x (0.90)	3.02x (0.31)